

CS240 Algorithm Design and Analysis

Spring 2026

Problem Set 3

Due: 23:59, March 26, 2026

1. Please write your solutions in English.
2. Submit your solutions to **Gradescope**. When submitting, match your solutions to the problems correctly.
3. If you want to submit a handwritten version, scan it clearly.
4. Up to 2 late days are allowed for homework submissions. Assignments submitted within the first 24 hours after the deadline will incur a 10% penalty; those submitted within 48 hours will incur a 20% penalty. Submissions more than 48 hours late will not be accepted unless there is a valid reason, such as a medical or family emergency.
5. You are required to follow ShanghaiTech's academic honesty policies. You are allowed to discuss problems with other students, but you must write up your solutions by yourselves. You are not allowed to copy materials from other students or from online or published resources. Violating academic honesty can result in serious penalties.

Notice: When proving that a problem A is NP-complete, you need to strictly follow the steps below and explicitly state each part.

- Show that $A \in \text{NP}$.

That is, given a candidate solution, we can verify it in polynomial time.

- Choose a known NP-complete problem B .
- Give a polynomial-time reduction from B to A .

For any instance b of B , construct an instance a of A in polynomial time.

- Prove correctness of the reduction.

Show that

$$b \text{ is a YES-instance of } B \iff a \text{ is a YES-instance of } A.$$

Problem 1:

Given:

- A set of communication links

$$L = \{\ell_1, \ell_2, \dots, \ell_n\}.$$

- A set of interfering pairs

$$I = \{(\ell_{a_1}, \ell_{b_1}), (\ell_{a_2}, \ell_{b_2}), \dots, (\ell_{a_m}, \ell_{b_m})\} \subseteq L \times L,$$

where each pair $(\ell_i, \ell_j) \in I$ means that links ℓ_i and ℓ_j cannot be activated simultaneously.

- An integer k .

Question:

Does there exist a subset $L' \subseteq L$ such that:

1. $|L'| \geq k$, and
2. for every interfering pair $(\ell_i, \ell_j) \in I$, at most one of ℓ_i or ℓ_j belongs to L' ?

Prove that this problem is NP-complete.

Solution:

Problem 2:

ShanghaiTech is planning a night security patrol system on campus. The campus is modeled as an undirected graph $G = (V, E)$, where each vertex represents an area, and an edge between two vertices means that the two areas are directly connected.

If a patrol point is placed in an area, then that patrol point can monitor both the area itself and all areas directly connected to it.

The university would like to place at most k patrol points so that every area on campus is monitored.

Formally, given an undirected graph $G = (V, E)$ and a positive integer k , decide whether there exists a subset $S \subseteq V$ such that

- $|S| \leq k$, and
- for every vertex $v \in V$, either $v \in S$, or there exists some $u \in S$ such that $(u, v) \in E$.

Prove that this decision problem is NP-complete.

Solution:

Problem 3:

Monotone 3-SAT: A *3-SAT formula* is a Boolean formula in which each clause contains exactly three literals. A clause is called *monotone* if all of its literals are positive or all of its literals are negative. For example, $(x \vee y \vee z)$ and $(\neg x \vee \neg y \vee \neg z)$ are monotone clauses, while $(x \vee y \vee \neg z)$ is not. As usual, a clause is satisfied if at least one of its literals evaluates to true.

Given a 3-SAT formula ϕ in which every clause of ϕ is monotone, decide whether ϕ has a satisfying assignment. Prove that MONOTONE 3-SAT is NP-complete.

Solution:

Problem 4:

Given a finite set of items I , where each item $i \in I$ has a positive integer size $s(i) \in \mathbb{Z}^+$, and two integer bin capacities B_1 and B_2 . Is there a way to partition the set I into two disjoint subsets I_1 and I_2 such that every item belongs to exactly one subset, and

$$\sum_{i \in I_1} s(i) \leq B_1 \quad \text{and} \quad \sum_{i \in I_2} s(i) \leq B_2?$$

Prove this problem is NP-complete.

Hint: You may use a reduction from the NP-complete problem Subset Sum.

Solution:

Problem 5:

Consider an $m \times n$ grid. Each cell is either empty, contains a black piece, or contains a white piece. You are allowed to remove some of the pieces. The question is whether it is possible to obtain a configuration such that

- each row contains at least one piece, and
- no column contains both black and white pieces.

Prove that this problem is NP-complete.

Solution: